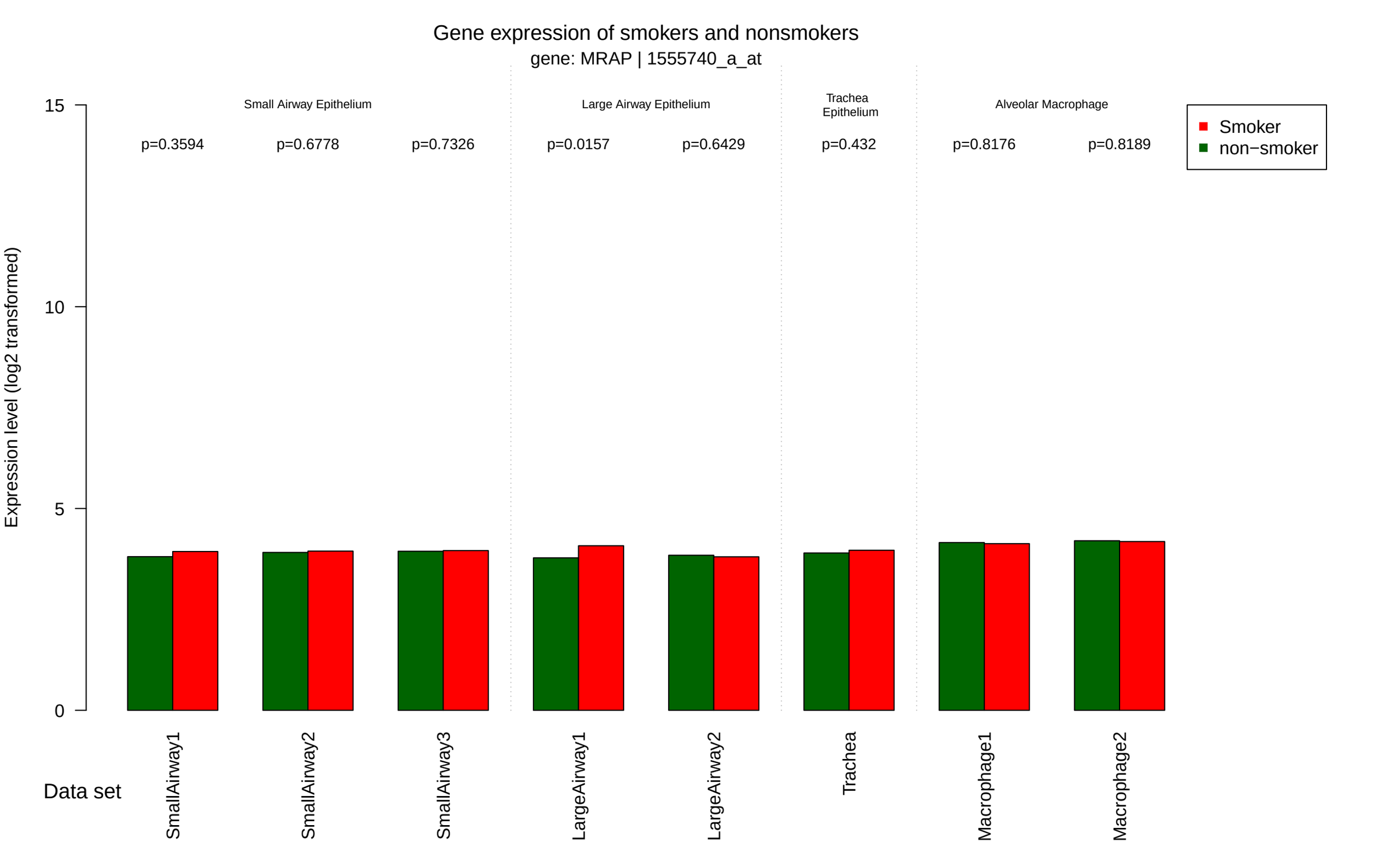
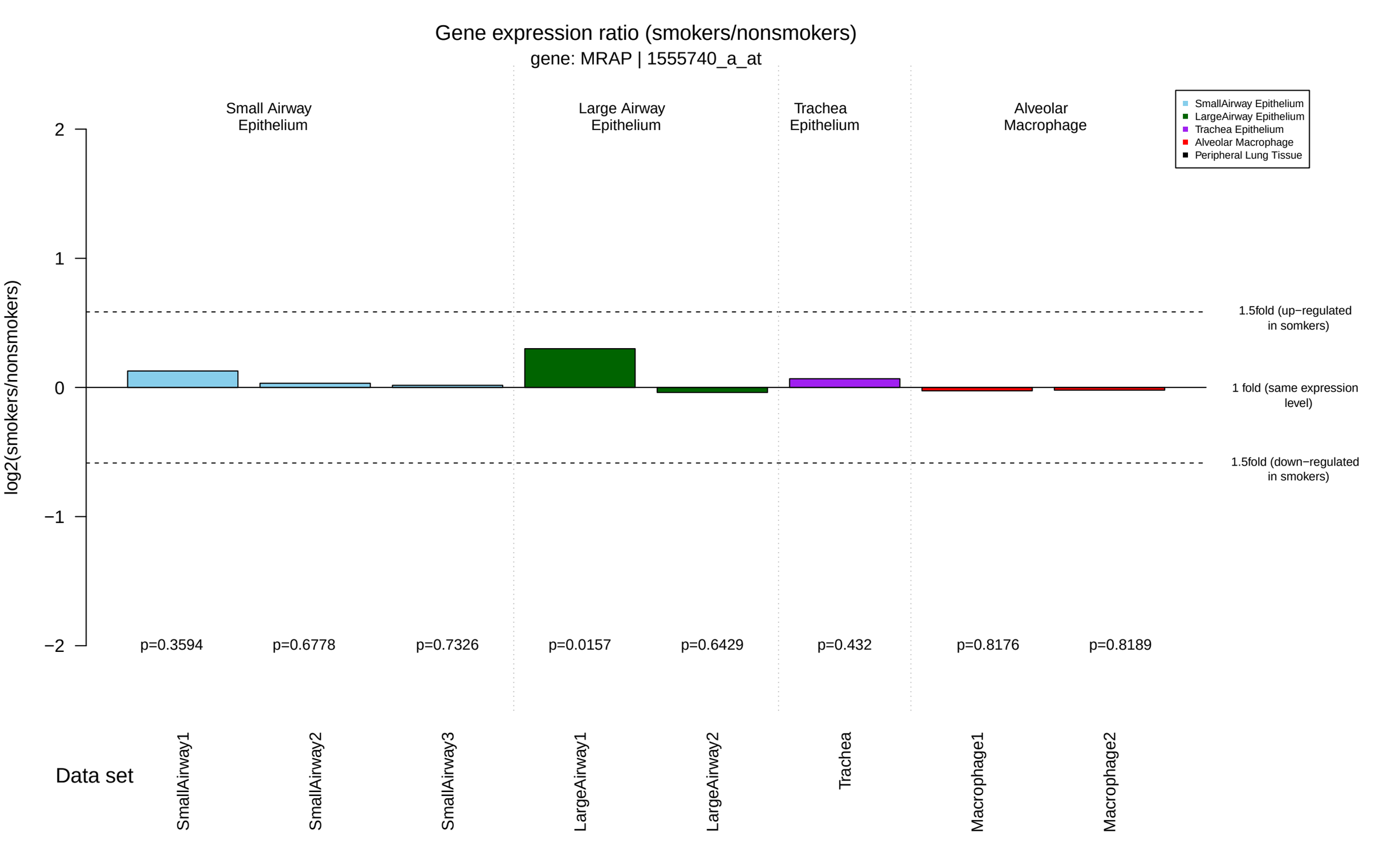
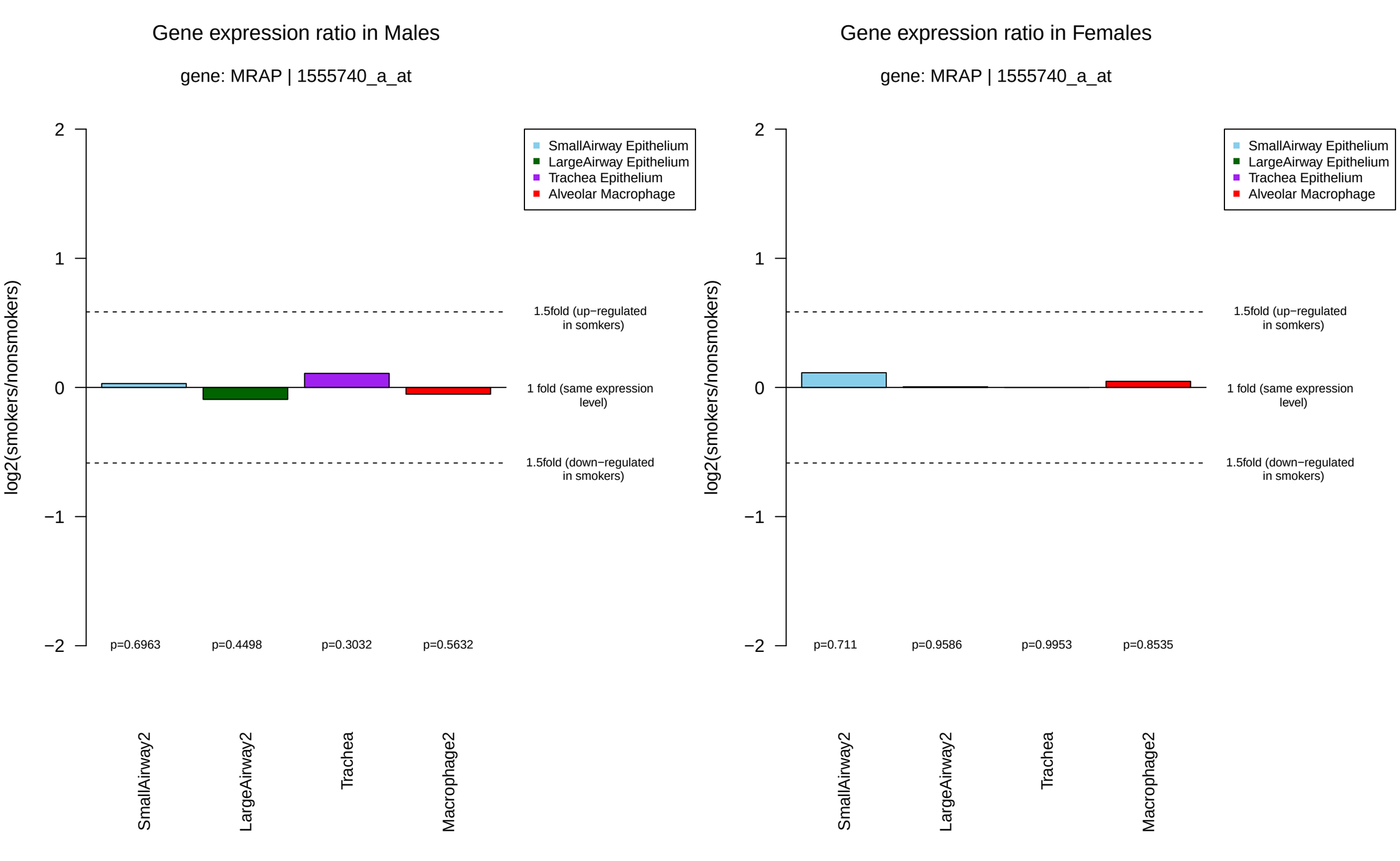
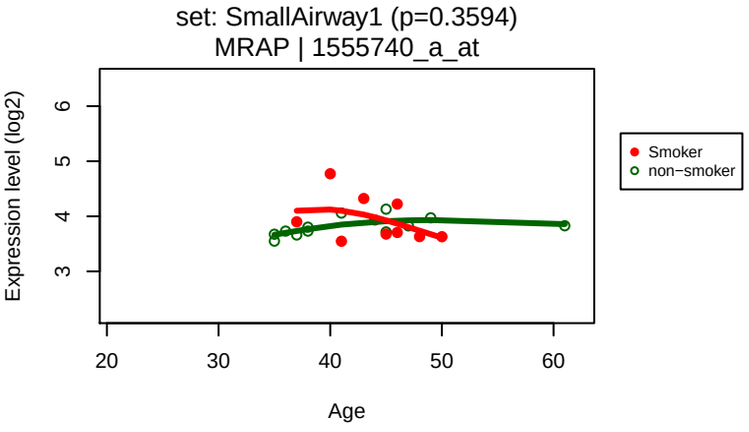
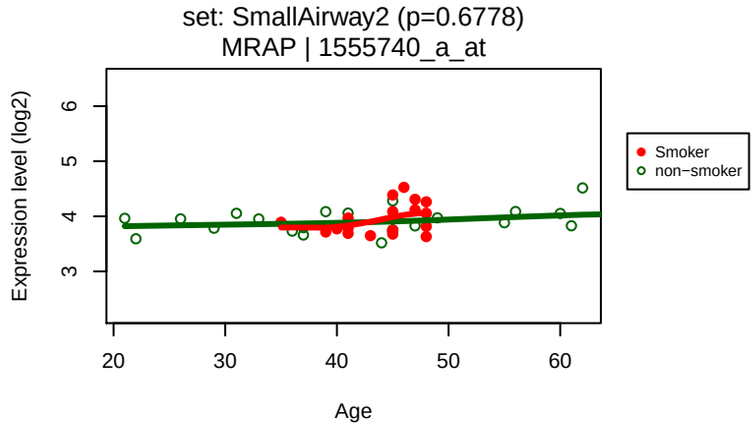
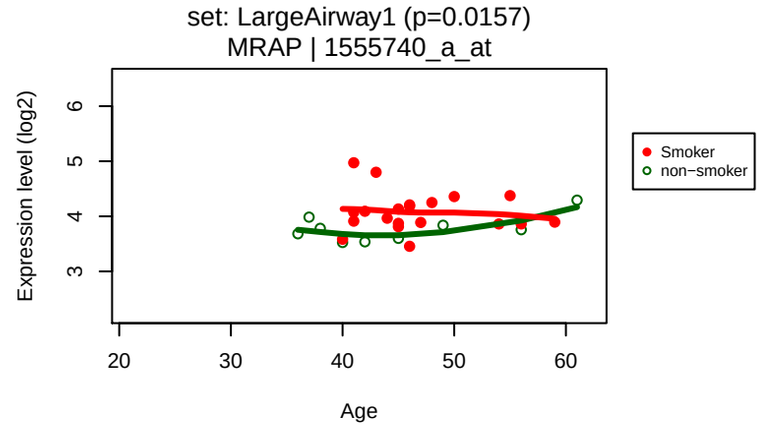

Gene expression of smokers and nonsmokers
gene: MRAP | 1555740_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3594
p=0.6778
p=0.7326
p=0.0157
p=0.6429
p=0.432
p=0.8176
p=0.8189
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

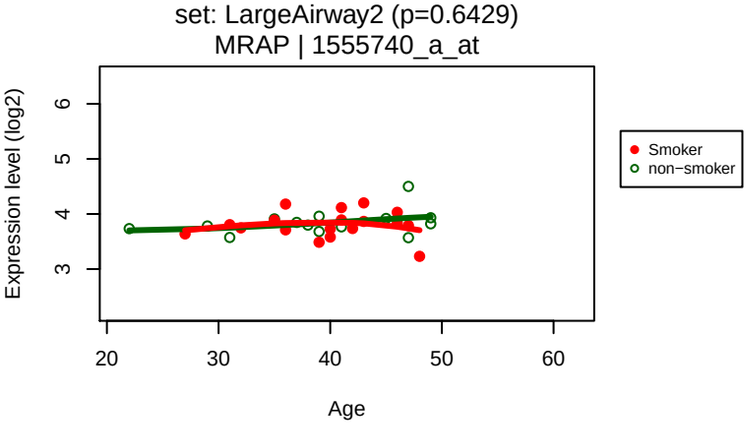

Gene expression ratio (smokers/nonsmokers)
gene: MRAP | 1555740_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3594
p=0.6778
p=0.7326
p=0.0157
p=0.6429
p=0.432
p=0.8176
p=0.8189
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

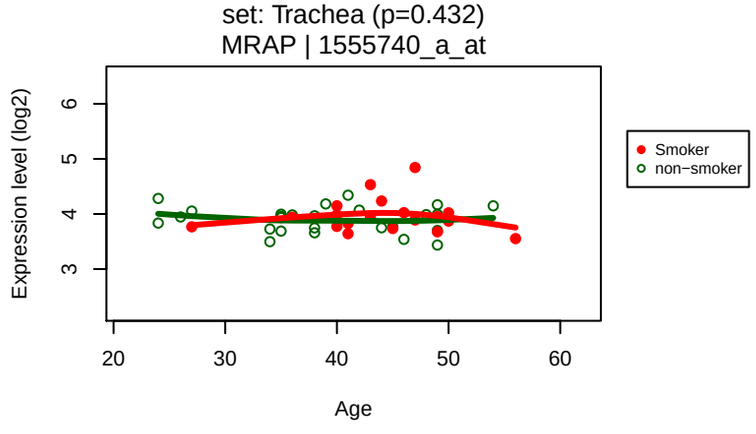

Gene expression ratio in Males
gene: MRAP | 1555740_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6963
p=0.4498
p=0.3032
p=0.5632
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRAP | 1555740_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.711
p=0.9586
p=0.9953
p=0.8535
SmallAirway2
LargeAirway2
Trachea
Macrophage2

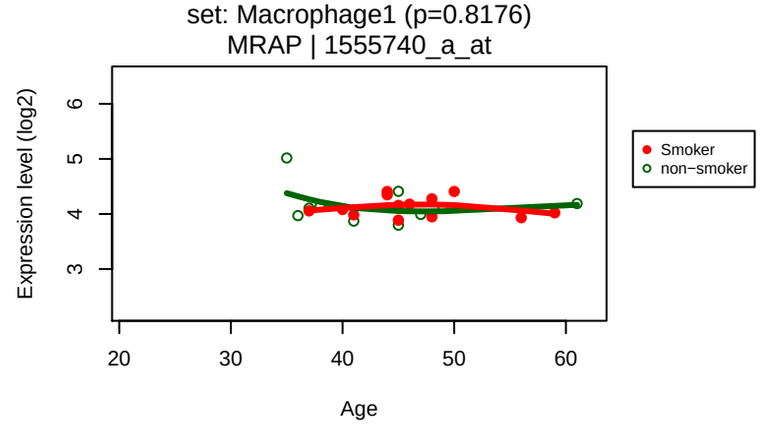

set: SmallAirway1 (p=0.3594)
MRAP | 1555740_a_at
Expression level (log2)
Age
Smoker
non−smoker

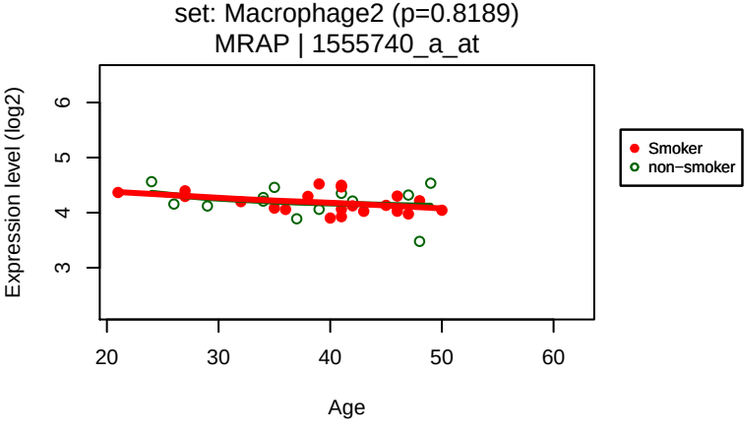

set: SmallAirway2 (p=0.6778)
MRAP | 1555740_a_at
Expression level (log2)
Age
Smoker
non−smoker

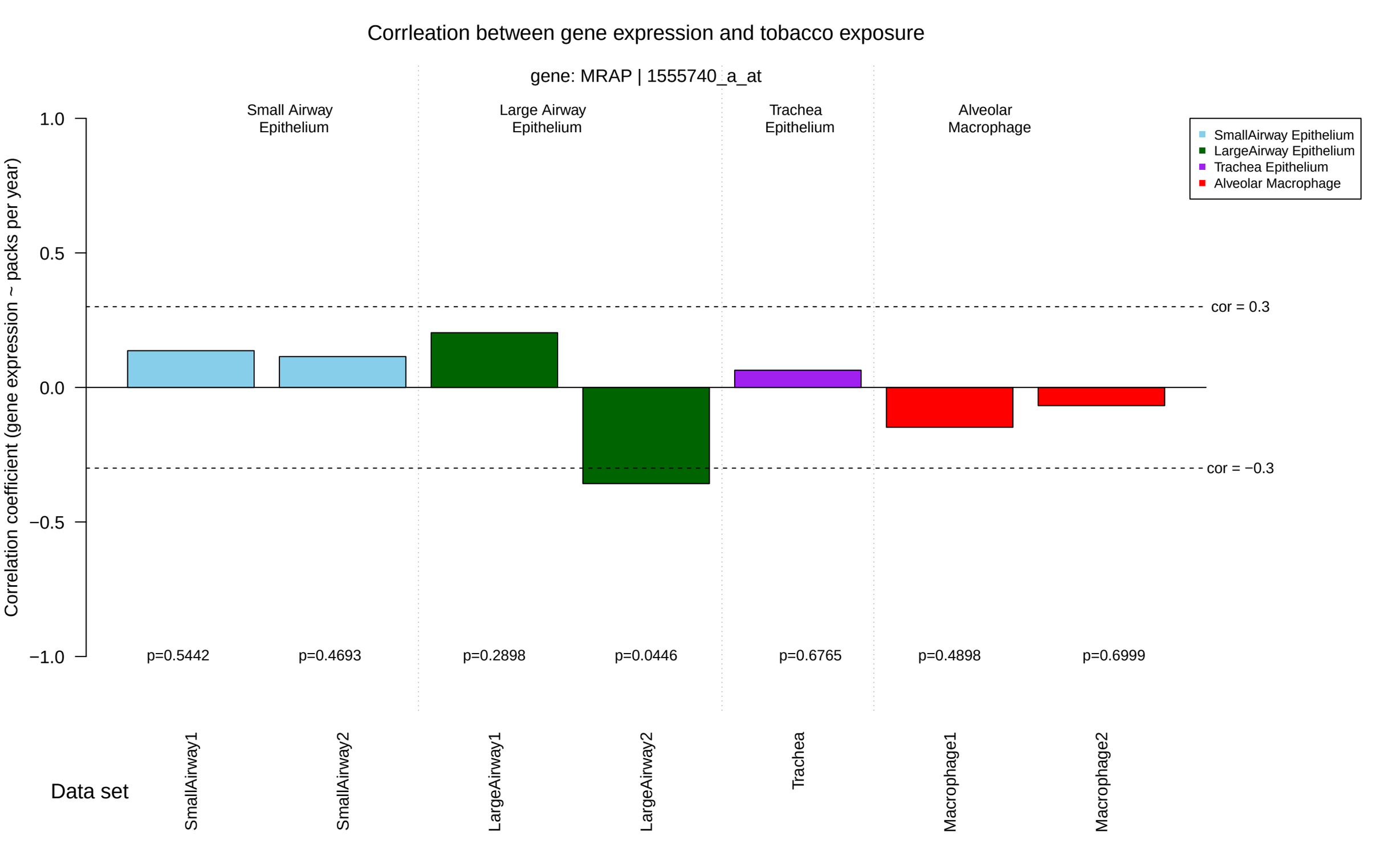

set: LargeAirway1 (p=0.0157)
MRAP | 1555740_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6429)
MRAP | 1555740_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.432)
MRAP | 1555740_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8176)
MRAP | 1555740_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8189)
MRAP | 1555740_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRAP | 1555740_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5442
p=0.4693
p=0.2898
p=0.0446
p=0.6765
p=0.4898
p=0.6999
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2